Kulachart Parnduangkeaw

Machine Learning & Data Science · Full-Stack Web & Mobile · Computer Engineering, SWU



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Top 30 KBTG KAMPUS Hackathon 2026

Top 24 NTT DATA Digital Innovation Challenge 2026

0.89 AUC-ROC skin-cancer classifier, 2026

~300 users SWU Run web app, 2026

Summary

Computer Engineering student at SWU (expected 2028) and 2026 Data Science Intern at Botnoi Group who builds ML systems end to end, from model to app. Seeking an AI / ML, Data Science or Full-Stack internship.

Experience

Data Science Intern — Botnoi Group · May – Jul 2026

- Built product demand-forecasting models for SME clients, including feature engineering that improved model accuracy.
- Set up the MLOps pipeline to train and deploy the models, and built dashboards for the client companies.

Projects

KaTang Cash-flow Co-pilot — KBTG KAMPUS Hackathon 2026 · Team of 3 · Data science · AI

- Built the gradient-boosting spending forecast and a Monte Carlo simulation of 4,000 thirty-day cash-flow paths (simulated data) behind a safe daily spending limit for variable-income first jobbers; pitched in the top-30 round.

  Python · scikit-learn · Pandas · Machine Learning

WELLYNC — NTT DATA Digital Innovation Challenge 2026 · Team of 3 · Web & LLM dev · AI · Web

- Built an Amazon Bedrock LLM agent with role-scoped tool calling that drafts chronic-care plans, saved only after a doctor confirms, on serverless AWS (Lambda, DynamoDB, Cognito); reached the top 24 teams.
- Built the Next.js web app where patients, caregivers and doctors share one consent-checked record.

  Next.js · TypeScript · Tailwind CSS · AWS

Derma AI: Skin Cancer Screening — CPE310 Healthcare AI · Solo project · 2026 · AI · Web

- Reached 0.89 AUC-ROC and 88.7% top-2 accuracy on 3,619 unseen test images with the best of the three models; exported to ONNX so it runs in the browser and images never leave the device.
- Fine-tuned ImageNet-pretrained ResNet50, EfficientNet-B3 and DenseNet121 on ISIC 2019 (25,331 images, 8 classes) with class-weighted loss for the imbalanced classes.
- Split by lesion (StratifiedGroupKFold) so no test lesion is seen in training; tuned each model's cancer threshold on validation to target 90% sensitivity.

  Python · PyTorch · scikit-learn · Pandas

Peak Electricity Day Alert — CPE312 Data Science capstone · 2026 · Data

- Forecast a week ahead which days hit peak demand from 2015–2020 Panama load and weather data, with leakage-safe daily features and a time-based train/test split.
- XGBoost reached **84% recall on peak days vs 73%** for a same-day-last-week baseline, at higher precision (78% vs 73%); also compared Logistic Regression.

  Python · XGBoost · scikit-learn · Pandas

More projects

Smart Flood AI — DOH Hackathon 2026 · Team of 3 · Web & AI dev · AI · Web. Zero-shot Grounding DINO pipeline that finds cars and wheels in highway CCTV to estimate water depth, plus the Next.js dashboard.

SWU Run Together 2026 — SWU Athletics Club · Team of 2 · Developer · 2026 · Web. React web app on live Google Sheets that gives runners a digital BIB with their order status; **used by about 300 runners**.

BrainFit — Client project · Team of 5 · 2025 · Mobile. Built the sign-in flow (Apps Script + Google Sheets) and some screens of a React Native (Expo) brain-training app commissioned by nursing students; **delivered to the client for their research**.

Employment Data Gap Analysis — Research project · 2025 · Data. Random Forest on 100,000 placement records vs a survey of 80 CPE students: students over-rated internships and portfolio (**31% vs 24%** model weight).

Prim's Algorithm Visualizer — CPE101 course project · 2024 · Web. D3.js tool that builds a Minimum Spanning Tree step by step.

This Portfolio — Web App · 2026 · Web. One typed data file drives kulac.dev and this CV, checked in CI on every push.

Leadership & Teaching

Computer Engineering Student Leader — Dept. of Computer Engineering, SWU · 2024 – Present

- Represent Computer Engineering students at faculty meetings and university events as the main link between students and faculty.
- Plan and run department workshops and student-development programs.
- Taught micro:bit programming to high-school students at a faculty activity and emceed a university event (2025).

Robotics Teaching Assistant — Srinakharinwirot University · 2026

- Taught robotics and programming fundamentals to high-school students in hands-on sessions.
- Prepared and maintained the robotics equipment and ran live robot-programming demos.

Metaverse Presentation Assistant — Srinakharinwirot University · 2025

- Assisted with a lecture on Metaverse technology (how VR, AR and AI work together in virtual worlds) and built its interactive slides and live demos.

Research Presentation Assistant — Dept. of Computer Engineering, SWU · 2025

- Helped a faculty member present a research poster at Chulachomklao Royal Military Academy on classifying Acanthamoeba cysts in microscope images with pretrained CNN features and an SVM classifier.

Education

Bachelor of Engineering in Computer Engineering — Srinakharinwirot University (SWU) · 2024 – Present · Expected 2028

Relevant coursework: Software Development, Database Systems, Artificial Intelligence, Healthcare AI Systems, Computer Architecture, Electronics

Honors & Certifications

- Top 30, KBTG KAMPUS Hackathon 2026 (KaTang Cash-flow Co-pilot)
- Top 24, NTT DATA Digital Innovation Challenge 2026 (WELLYNC)
- Bangkok Insurance Scholarship — Bangkok Insurance, 2025
- Google Cloud, Big Data & GenAI Training — Google Cloud, 2026

Skills

AI / ML: Machine Learning, Deep Learning, Computer Vision, Generative AI, PyTorch, scikit-learn, XGBoost, OpenCV, Grounding DINO, ONNX Runtime, Amazon Bedrock (LLM agents, tool calling)

Methods: Demand forecasting, time-series forecasting, transfer learning, time-based validation, threshold tuning, Monte Carlo simulation, MLOps

Programming: Python, SQL (MySQL), Java, C/C++

Data: NumPy, Pandas, Data Analysis, Matplotlib, Seaborn, Google Sheets, Google Colab

Web: React, Next.js, TypeScript, JavaScript, Tailwind CSS, D3.js, HTML/CSS, Vite, Framer Motion

Mobile: React Native, Expo

Backend & cloud: FastAPI, AWS (Lambda, API Gateway, DynamoDB, Cognito), Firebase, Google Apps Script, Google Cloud

Tools: Git, GitHub Actions, Vercel, Hugging Face Spaces

Languages: Thai (native), English (intermediate)

Scan for live demos & project write-ups

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